25900896

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE

EXAMINATION NOVEMBER 2025 FIRST SEMESTER (2024 ADMISSION ONWARDS)

DSC - MG1DSCMBG100 - UNSEEN WORLD OF MICROBES

Duration: 2 Hours Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Part A – Short Answer - Answer Any 10 out of 12 – 2Marks 2X10=20

1. Mention any two scientists who contributed to the Golden era of Microbiology. [K] / [CO1] Louis Pasteur and Robert Koch. **[1+1]**
2. What is mRNA therapy? [K] / [CO2] - A treatment that uses messenger RNA to instruct cells to make a specific protein to fight a disease. **[2]**
3. Write notes on Protozoa? [U] / [CO3] - Unicellular, eukaryotic, heterotrophic organisms, often found in water/soil; some cause disease. Any 2 points **[2]**
4. Mention the characteristics of organisms classified under the kingdom Animalia. [K] / [CO2] - Multicellular, eukaryotic, heterotrophic organisms that lack cell walls. **[2]**
5. What are some examples of beneficial microbes found in the human body, and what roles do they play? [U] / [CO3] - *E. coli* in the gut (aids digestion) and *S. epidermidis* on the skin (protects from harmful bacteria). Any 2 organisms **[1+1]**
6. Name the contribution of Francesco Redi [K] / [CO1] - He experimented with jars and meat to disprove spontaneous generation. **[2]**
7. What was the first antibiotic discovered? [K] / [CO1] - Penicillin. **[2]**
8. How does the domain Bacteria differ from Archaea? [K] / [CO2] – Any 2 difference **[1+1]**
9. What is mRNA therapy? [K] / [CO2] - A treatment that uses messenger RNA to instruct cells to make a specific protein to fight a disease. **[2]**
10. What is monochrome staining [U] / [CO3] - Staining with a single dye to highlight basic cell shapes and arrangements. **[2]**
11. Give an example for an acidic stain and a basic stain [U] / [CO3] - Acidic stain: Nigrosin. Basic stain: Crystal violet. **[1+1]**
12. Cite the roles of a medical microbiologist [K] / [CO1]- Identify pathogens and recommend treatments to diagnose and combat infectious diseases. **[2]**

Part B – Short Essay - Answer Any 4 out of 6 – 5Marks 5X4=20

13. Write a note on Linnaeus's 2-kingdom classification? [K] / [CO2] - Proposed by Carolus Linnaeus. **[1].** Divided life into **Kingdom Plantae** (plants, autotrophs) and **Kingdom Animalia** (animals, heterotrophs). **[3]. Limitation:** Did not distinguish between prokaryotes/eukaryotes or unicellular/multicellular organisms. **[1]**
14. List five structural differences between prokaryotes and eukaryotes. [U] / [CO3] - **Nucleus:** Prokaryotes lack a true nucleus; Eukaryotes have a membrane-bound nucleus. **[1]. DNA:** Prokaryotes have circular DNA; Eukaryotes have linear chromosomes. **[1]. Organelles:** Prokaryotes lack membrane-bound organelles; Eukaryotes have them **[1]. Wall:** Prokaryotes have peptidoglycan; Eukaryotes have cellulose or chitin **[1]. Cell Size:** Prokaryotes are generally smaller than Eukaryotes **[1]**
15. Write notes on Archaebacteria? [U] / [CO3] - A primitive group of prokaryotes. **[1]** **Unique Trait:** Cell walls lack peptidoglycan and have unique membrane lipids. **[2]** **Habitat:** Often live in extreme environments (e.g., hot springs, salt lakes). **[1].** Any 1 example **[1]**
16. Identify the beneficial and harmful roles of microorganisms in the food and dairy industry. [K] / [CO1] - **Beneficial (3 marks):** Fermentation: Production of yoghurt, cheese, bread. **[2]** Probiotics: Beneficial bacteria for gut health. **[1]. Harmful (2 marks):** Spoilage: Causes food to rot and smell. **[1].** Food Poisoning: Pathogens like *Salmonella* cause illness. **[1]**
17. Describe the biological sources of antimicrobial peptides. [U] / [CO2] - AMPs are natural defence molecules produced by most living organisms. **[1].** Sources: **[2**]. Example **[2]**
18. Relate medical microbiology to public health. [K] / [CO1] - Medical microbiology identifies pathogens causing disease. **[2]** This information is used for public health actions, such as tracking outbreaks, recommending vaccines, and advising on sanitation to prevent the spread of infection within the community. **[3]** Any 5 points.

Part B – Long Essay - Answer Any 1 out of 2 – 10 Marks 10X1=10

19. State Biogenesis vs Abiogenesis. Mention the experiments supporting both theories. [K] / [CO1] –
    Abiogenesis (Theory of Spontaneous Generation) **(4 marks):** - Definition **[1]**. Historical Support & Experiments: Common Belief: (maggots appeared on rotting meat) **[1].** Supporting Experiment - John Needham's Experiment: Boiled broth in a flask, sealed it, and it became cloudy, claiming it proved spontaneous generation. **[2].**
    Biogenesis **(6 marks):** Definition: **[1].** Supporting Experiments**:** Francesco Redi's Experiment **[2.5].** Louis Pasteur's Experiment**:** Used swan-neck flasks. **[2.5].** **[10]**

20. What are the main characteristics and features of Viruses? – Introduction **[2]** Any 8 points **[8].** Acellular Structure, Genetic Material, Protein Coat, Obligate Parasites, Host Specificity: Infect specific types of host cells, Size, Shape, Envelope, Remain inactive outside the host, Examples. **[10]**